

Quantifying Potential Recharge in Mantled Sinkholes Using ERT

by Benjamin F. Schwartz^{1,2} and Madeline E. Schreiber¹

Abstract

Potential recharge through thick soils in mantled sinkholes was quantified using differential electrical resistivity tomography (ERT). Conversion of time series two-dimensional (2D) ERT profiles into 2D volumetric water content profiles using a numerically optimized form of Archie's law allowed us to monitor temporal changes in water content in soil profiles up to 9 m in depth. Combining Penman-Monteith daily potential evapotranspiration (PET) and daily precipitation data with potential recharge calculations for three sinkhole transects indicates that potential recharge occurred only during brief intervals over the study period and ranged from 19% to 31% of cumulative precipitation. Spatial analysis of ERT-derived water content showed that infiltration occurred both on sinkhole flanks and in sinkhole bottoms. Results also demonstrate that mantled sinkholes can act as regions of both rapid and slow recharge. Rapid recharge is likely the result of flow through macropores (such as root casts and thin gravel layers), while slow recharge is the result of unsaturated flow through fine-grained sediments. In addition to developing a new method for quantifying potential recharge at the field scale in unsaturated conditions, we show that mantled sinkholes are an important component of storage in a karst system.

Introduction

Recharge and infiltration are important hydrologic processes that are difficult to understand and quantify at the field scale, both in terms of rates and volumes. The process of water movement from the surface into the subsurface is defined as infiltration, while recharge is defined as water that is added to an aquifer (Scanlon et al. 2002). Potential recharge refers to water that has infiltrated and drained to a depth at which it may be assumed to recharge an aquifer at some time in the future (Scanlon et al. 2002). Exceptions to this assumption are in arid regions or in areas with extremely deep unsaturated zones, in which even deep drainage may not become recharge. Understanding where, when, and how

much water recharges an aquifer is critical information for understanding ground water quality and quantity (de Vries and Simmers 2002; Gunn 1986).

In many settings, recharge is a relatively slow process resulting from diffuse infiltration of precipitation, followed by unsaturated flow through soils, sediment, and bedrock, until a portion of the water (the deep drainage component) reaches the water table. One notable exception to this slow process is in karst regions where a significant portion of the recharge entering an aquifer can occur directly via streams sinking into conduits. While this process, called rapid or point source recharge, can quickly add large volumes of water to a karst aquifer, an extension of the same conduit system may also rapidly transport and remove it and residence time can be very short. Rapid recharge is often cited as one reason why karst aquifers are extremely sensitive to contamination, and sinkholes, in particular, are targets of concern as a potential source of significant contamination (Lee and Krothe 2001; Stephenson et al. 1999; Tranter et al. 1997).

The assumption that sinkholes provide both direct connections and pathways for rapid flow and transport between the surface and the underlying karst aquifer may

¹Department of Geosciences, 4044 Derring Hall, Virginia Polytechnic and State University, Blacksburg, VA 24060.

²Corresponding author: Currently at Department of Biology, Texas State University, 601 University Dr., San Marcos, TX 78666; 1-512-245-7608; fax: 1-512-245-8713; bs37@txstate.edu

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not be entirely accurate in sinkholes containing thick clay- and silt-rich soils. These soils have the capacity to store and slowly release large amounts of water, in addition to allowing some water to pass relatively rapidly through the unsaturated zone via macropores such as old root casts, burrows, and soil fractures (McKay et al. 1993; Iqbal and Krothe 1995). Adding to the complexity, runoff after heavy precipitation can be funneled to the bottom of the sinkhole where it is forced to infiltrate under ponded conditions. Even small sinkholes may capture runoff from very large areas and have the potential to contribute recharge volumes disproportionate to their size. Thus, thick soils in the unsaturated zone over a karst aquifer can be a significant source of slowly released water to a karst hydrologic system. The diversity of infiltration processes in mantled sinkholes ultimately means that these sinkholes have the potential to provide recharge to an aquifer at varying rates (Gunn 1986).

Many methods have been developed for estimating or measuring recharge at different spatial scales (Scanlon et al. 2002). For large-scale studies, recharge can be modeled using mass-balance models and pedotransfer functions, which may incorporate data such as soil water, soil properties, stream and spring hydrographs, well levels, precipitation, stable isotopes, and potential evapotranspiration (Sophocleous 1991; Das Gupta and Paudyal 1988; Doctor et al. 2006; Nolan et al. 2007). Geographic information systems are also useful tools for regional recharge modeling, especially for use with readily available data such as soil types, elevation, land cover, and climatic parameters (Dripps and Bradbury 2007; Arnold et al. 1998; Srinivasan et al. 1998). At smaller scales, tools such as lysimeters provide good estimates of recharge, though they are subject to large spatial variations in estimated recharge due to localized heterogeneities in soils and vegetation (Chapman and Malone 2002).

Unfortunately, quantifying amounts and rates of infiltration and recharge at intermediate scales (tens to hundreds of meters), typical of many field sites, is extremely difficult. Intermediate-scale recharge can be estimated using physical measurements obtained at a small scale (e.g., lysimeters or soil moisture sensors) and then applied to a larger area of interest through interpolation or extrapolation. However, as aforementioned, physical and hydrologic parameters are heterogeneous at the small scale and recharge estimates derived at this scale may significantly over- or underestimate recharge at intermediate-scale field sites. Thus, to quantify infiltration and recharge at the intermediate scale, it is better to make measurements that are relevant at that scale. Geophysical methods are excellent for this application because of their scalability and mobility, and because measurements are more likely to represent an average parameter value for the area or volume of interest. Electrical techniques such as electrical resistivity tomography (ERT) are particularly well suited for work in the unsaturated zone because of their ability to penetrate to useful depths with reasonable resolution in the field and their high sensitivity to

changes in electrical properties resulting from changes in water content (Sreedeeep and Singh 2005; Michot et al. 2003; Sheets and Hendrickx 1995). The primary objectives of this study were to quantify variations in soil water content at the field scale, and to determine the timing and volume of infiltration and potential recharge occurring in sinkholes with thick soil mantles after a rainy period. We accomplished these objectives using differential ERT-derived changes in water content over a period of approximately 5 months (May 17 to October 9, 2006).

Field Site

Our research site at the Virginia Tech Kentland Experimental Farms in Montgomery County, Virginia, contains two well-developed sinkhole plains formed in New River terraces (Figure 1). Sinkholes are generally broad and shallow, allowing easy access for agricultural activities, and contain no bedrock outcrops and rare evidence of minor subsidence or soil piping. Thick terrace deposits mantle sinkholes with soils characterized as weathered fluvial terrace materials deposited by the New River, which have developed over the underlying Cambrian-aged Elbrook Formation limestone and dolostone bedrocks. Soils are classified by the U.S. Department of Agriculture, Natural Resources Conservation Service (USDA-NRCS) as Guernsey silt loam, Unison and Braddock soils, and Unison and Braddock cobbly soils (USDA-NRCS 2006). Both sinkhole plains have numerous sinkholes of similar size and shape. Six sinkholes were initially chosen as being representative of most sinkholes in both plains and because they were accessible for research. They were then characterized with topographic surveys and more than 100 ERT profiles. Of these six sinkholes, two were chosen for more detailed analysis in this study because they are similar in size, easily accessible, and installation of instrumentation would not interfere with farming activities. Sinkhole #1 (S1) is in a higher and older terrace deposit and contains highly weathered soils to depths exceeding 12.2 m. Sinkhole #5 (S5) is formed in a lower and younger terrace and contains soils that are not as mature. In S5, bedrock was reached in most augered holes at depths between 3.4 and 7.6 m below land surface. Instrumentation installed at the field site consists of time domain reflectometry (TDR) access tubes used to obtain small-scale volumetric water content measurements and permanent carbon electrode arrays for ERT measurements.

Sinkhole formation at both sites appears to be the result of two mechanisms. First, rare soil piping, downslope movement, and slumping are slowly moving soils from the surface into the subsurface. A second, and perhaps more important, mechanism is the dissolution of bedrock and subsequent slumping of sediments into the resulting bedrock depression. Evidence for this can be found in cobble layers and other soil layers that are laterally continuous in sinkhole flanks but slope toward the sinkhole bottoms and become deeper below the surface

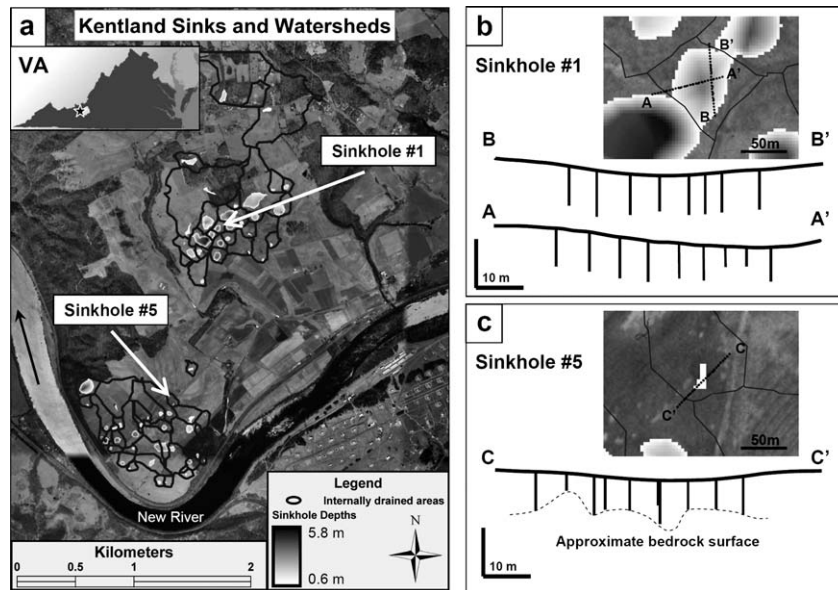


Figure 1. Field site and locations of ERT profiles: (a) Virginia Tech Kentland Experimental Farms at Whitethorne, Virginia; (a) shows study sinkholes S1 and S5 and catchment areas (adjacent polygons) for each sinkhole within the two sinkhole plains. Aerial imagery © 2002 Commonwealth of Virginia. S1 is in the higher, older terrace. (b) and (c) show the location and orientation of instrumentation installed in transects across both sinkholes. Upper image in the diagrams is a map view of the sinkhole, while the lower portion of the diagrams shows profile views of TDR access tubes and other instrumentation installed along each transect. Twenty-five ERT electrodes are spaced evenly along each 72-m transect (A to A', B to B', and C to C'). Note that depth of bedrock was not determined in S1.

with proximity to the sinkhole bottom. Vegetation at the site is dominated by grasses. In the upper S1, fields are used for hay and grazing. During the study period, grass was either mowed or grazed. In the lower S5, the site was covered by grasses and weeds that were kept mowed to less than 50 cm in height.

Methods

Our approach to quantifying potential recharge in mantled sinkholes included measuring temporal variations in volumetric water content derived from differential ERT data. In soils, potential recharge occurs when the rate of precipitation exceeds the rate of potential evapotranspiration (PET) over a time period sufficient for infiltrating water to drain below root depth. By assuming a 1.5-m maximum root depth for the fescue and orchard-grass, which dominate the field site (Ash et al. 1975; Ridley and Simpson 1994), and further assuming that water that moved below this depth would no longer be influenced by evapotranspiration (ET), we calculated potential recharge after a series of large storm events by measuring temporal changes in water content below the root zone (Figure 2). Cumulative changes in three ERT-derived water content profiles were combined with rates of cumulative precipitation and PET over the same study period to identify times when PET or infiltration and drainage dominated the system. Recharge as a percent of precipitation was calculated as the ratio of volumetric water increase below 1.5 m vs. volume of total precipitation.

Deriving Volumetric Water Content from ERT

Dipole-dipole ERT data were collected 11 times between May 17, 2006, and October 9, 2006, for each of the three transects across the two study sinkholes (Figure 1):

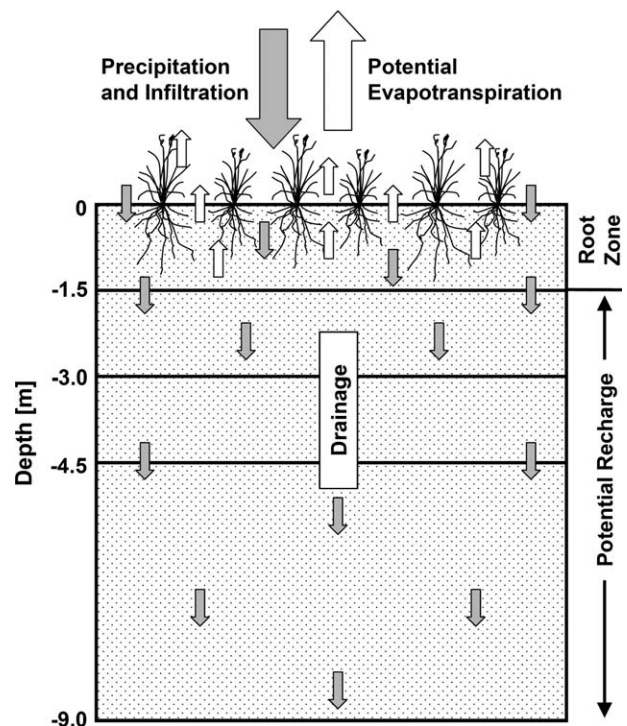


Figure 2. Conceptual diagram of the processes modeled and methods used to calculate potential recharge in unsaturated soil profiles.

two in S1 and one in S5. Data were collected using permanently installed arrays of 25 carbon ERT electrodes per transect, 72 m in length. We used permanent carbon electrodes, a technique used by Stevens (2007) for long-term monitoring of variable-density ground water flow in a saline environment, to ensure high-quality ERT data and to minimize errors that would have resulted from slight variations in soil-electrode electrical contact and location if we had installed and removed electrodes each time measurements were made.

Volumetric water content was calculated for the outlined portions of each ERT profile (Figure 3) by converting resistivity values into water content using a modified and numerically optimized form of Archie's law (Schwartz et al. 2008a; Shah and Singh 2005) as shown in the following:

$$\theta = (\sigma_b / c \cdot \sigma_w)^{1/m} \quad (1)$$

where σ_b is the bulk conductivity (1/ERT-derived resistivity), σ_w is the pore water conductivity, θ is the volumetric water content, and c and m are the fitting parameters that depend primarily on the particle size characteristics of the soil:

$$c = 0.00873 \text{Clay}^{2.608} \quad (2)$$

$$m = 0.4848 \text{Clay}^{0.08183} \quad (3)$$

where clay is defined as the percent by weight of a soil sample with a particle size fraction 2.0 μm or less. The model includes the important effects of clay content on bulk soil conductivity, and a Mehlich 1 extractable Ca + Mg proxy for pore water conductivity (Shah and Singh 2005; Schwartz et al. 2008a). The proxy assumes a relationship between the dominant cation species in the soil and equilibrium pore water conductivity. Resistivity values used in the model were derived from the results of processing each two-dimensional (2D) dipole-dipole ERT profile using Advanced Geosciences Inc. EarthImager 2D (2007) software. Each profile was discretized into 0.5×0.5 m cells, with centered cell nodes representing the resistivity value for the cell.

Changes in water content for 10 time intervals from May 17 to October 9, 2006, were obtained by calculating the difference in ERT-derived volumetric water content in each model cell relative to the initial data set collected on May 17, 2006 (Julian d 137). These changes were calculated as cubic meters of water added or subtracted from each soil profile (Figure 3) by assuming a 1-m profile thickness. Because we calculated volumetric water content, conversion of these data into a change in volumetric water in the profile (relative to the initial profile) was done by sequentially summing the volumetric water changes for each of the 0.5×0.5 -m model cells contained in the 1-m-thick profile.

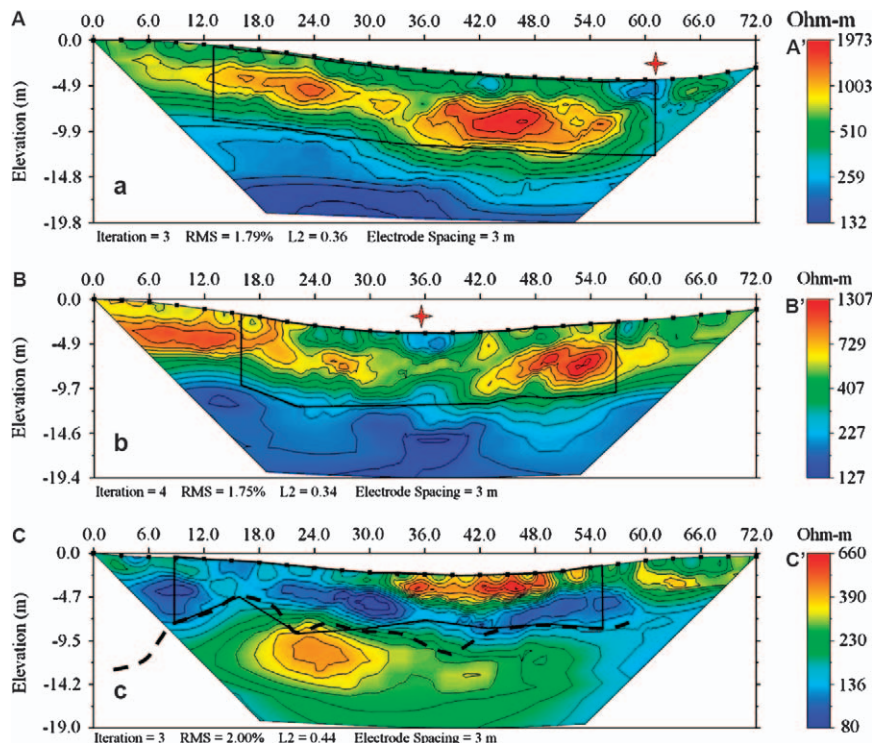


Figure 3. Representative ERT profiles for each transect collected on May 17, 2006, showing contoured resistivity values and portions of each profile where time series water content data were modeled (noted by boxed areas). Transects (a) and (b) are in S1, while transect (c) is in S5. The bedrock-soil interface in (c) is shown as a dashed line. Note that the bedrock-soil interface was not encountered in (a) or (b). Stars indicate the point where (a) and (b) intersect. A, A', B, B', C, and C' represent transect end points as shown in Figure 1. Also note that the resistivity scales are not the same for each profile.

PET Modeling

We used the American Society of Civil Engineers (ASCE) Penman-Monteith model (Allen et al. 1998; ASCE-EWRI (Environmental and Water Resources Institute) 2002; Howell and Evett 2004; Snyder and Eching 2007) with a tall grass reference to calculate daily PET at our field site. A spreadsheet model (PMday) written by Snyder and Eching (2007) was used to perform PET calculations. Details of the ASCE Penman-Monteith model (ASCE-PM) and how PET is calculated can be found in the aforementioned references. Required climate data were derived from hourly data recorded at our field site on the Virginia Tech Kentland Farms (2007) and include daily maximum and minimum temperature, average wind speed, global solar radiation (converted to net solar radiation), and daily maximum and minimum relative humidity. Hourly precipitation data were also recorded at this weather station. PET is a modeled estimate of the environmental demand for water, assuming ample water is available. Because ample water is often not available, annual PET is almost always greater than actual evapotranspiration (AET).

Cumulative PET and precipitation were both converted into cubic meters of water by multiplying the modeled or measured amount by each profile's length and assuming surface width of 1 m. It should be noted that AET was not directly modeled in this study. Except for periods when the rate of precipitation is greater than the rate of PET, PET was assumed to be greater than AET.

Potential Recharge Calculations, Assumptions, and Limitations

Potential recharge for the study period was calculated using a mass-balance approach, which assumes that changes in volumetric water content below -1.5 m represent changes in potential recharge. To measure the rate of drainage through each soil profile, we divided ERT-derived water content profiles into intervals of 0 to -1.5 m, -1.5 to -3.0 m, -3.0 to -4.5 m, and below -4.5 m in depth (Figure 2). This allowed us to monitor movement of wetting fronts and test whether water was draining through the profile faster than the measurement interval (days to weeks), which was critical for determining whether potential recharge is accurately measured using this method. As an example of why this is important, consider a situation where infiltrating water drains through the soil profile faster than can be detected with a certain monitoring interval. Data could incorrectly show that no significant increase in water content (potential recharge) occurred because the wetting front had already passed by the time the second measurement was made. Breaking the model into depth intervals also allowed a more detailed investigation of differences in drainage rates between the two sinkholes that could be related to differences in soil properties.

Assuming that the rate of drainage through the deeper portion of the profile requires a time period that is significantly longer than an individual measurement interval, potential recharge from a single wetting front can be calculated as the maximum amount of water that is added to the entire profile below -1.5 m. The

maximum amount can be used if we assume that no water will move upward after it drains below -1.5 m and that it represents the entire volume of potential recharge from the precipitation event monitored. Thus, decreases in water content after a maximum is reached can only represent drainage through the bottom of the profile.

We did not observe evidence of soil piping during our field studies or in the more than 100 ERT profiles conducted in six sinkholes at the site. However, there are verbal reports of rare soil piping occurring historically at the field site. Depending on the size, extent, and proximity to the surface of any potential soil pipes, we may not be able to detect soil pipes using ERT. If this is the case, soil pipes may contribute significantly to rapid recharge, which would complicate recharge calculations.

Results

Relationship between PET, Precipitation, and Soil Water Content

Figure 4 shows cumulative PET, precipitation, and ERT-derived water content for each profile (regions outlined in Figure 3) over the study period. Examination of the relationships between these data sets yields useful information regarding potential for recharge to occur. For example, when the rate of precipitation is less than the rate of PET (sections I and III in Figure 4), there is no potential for infiltration and drainage of water deeper than the root zone. Conversely, when the rate of precipitation is greater than the rate of PET (section II in Figure 4), potential recharge can occur when infiltrating water drains below the depth of influence from ET processes (-1.5 m), assuming soils are not frozen. Rates of precipitation and PET can be determined by examining the slopes of cumulative precipitation or PET curves. For example, when the slope of cumulative precipitation is equal to or steeper (more positive) than that of PET, potential recharge can occur.

Three important time intervals can be described in our data (Figure 4). The first is a period of drying from d 137 (May 17, 2006) until d 173 (June 22, 2006) (section I in Figure 4). During this period, there was very little precipitation and PET removed water from the soil faster than it was added by precipitation; thus, the soils near the surface were losing water. This trend was clearly observed in S1 (Figures 4a and 4b). S5 showed a slightly different response. Although the precipitation and PET data used are identical to those used for S1, water content in the entire profile remained essentially constant during this period, with a slight increase possibly related to small rain events. S5 contains soils with higher silt and lower clay content that allowed water to infiltrate and drain more rapidly than in the higher clay content soils found in S1 (Schwartz et al. 2008b).

On d 173 (June 22, 2006), a period of increased precipitation began in which the rate of cumulative precipitation was greater than or equal to the rate of cumulative PET (Figure 4, section II). These conditions persisted

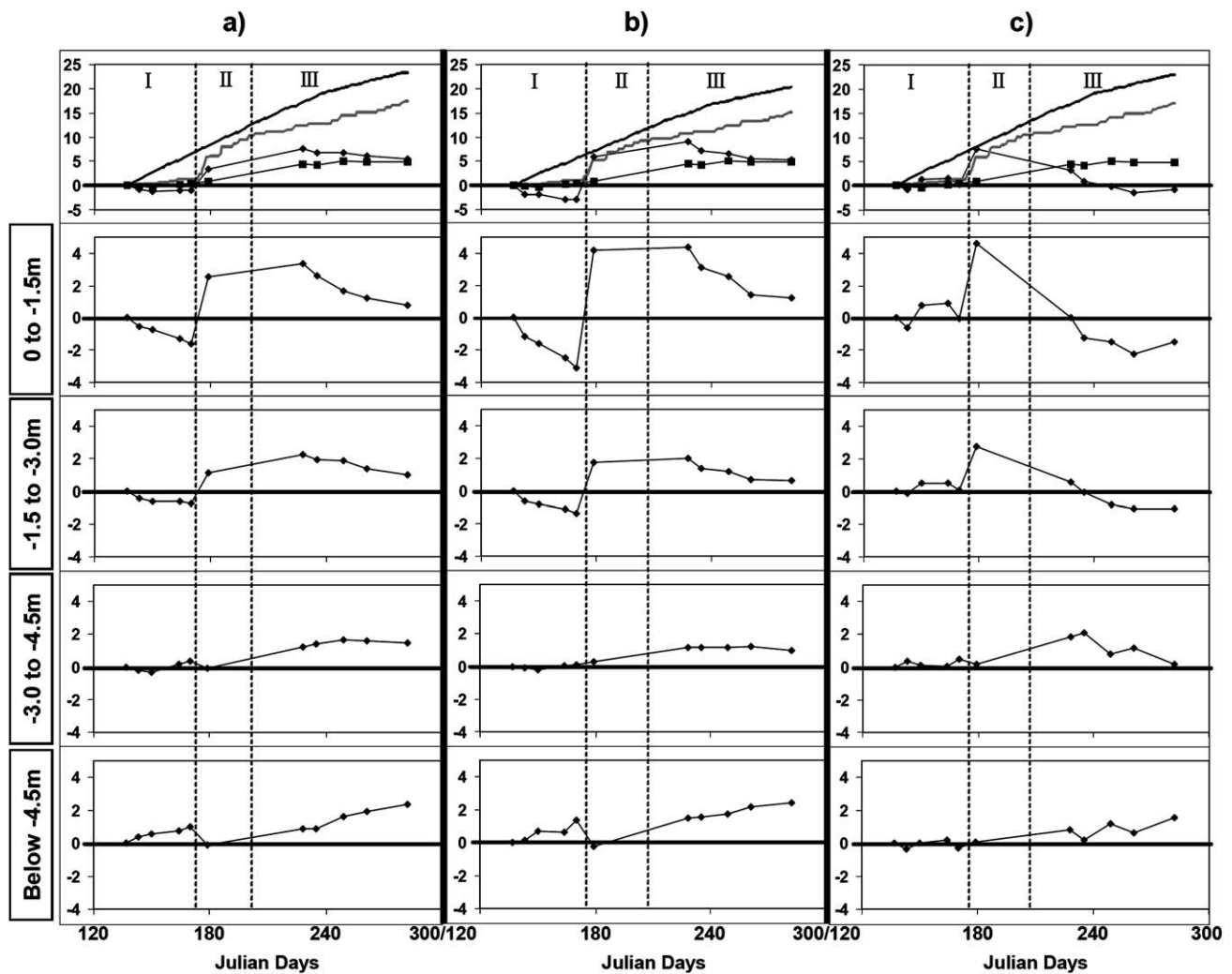


Figure 4. Top row shows cubic meter change in cumulative PET (top curve), cumulative precipitation (lower curve), cumulative change in total-profile soil water content (diamonds), and cumulative change in soil water content between -1.5 and -9.0 m (squares) in S1 (a and b) and S5 (c). Sections I and III denote periods of drying and section II denotes a period of wetting. Lower rows show cumulative changes in ERT-derived water content for each sinkhole transect by depth interval. Note the progression of a wetting front as detected by the time required for water to be added to sequentially deeper intervals in the profiles.

until d 204 (July 23, 2006) when the rate of precipitation decreased. In both sinkholes, measured water content increased sharply between d 170 (June 19, 2006) and d 179 (June 28, 2006). Major differences between the two sinkholes appear in the data collected on d 228 (August 16, 2008). In S1, water content continued to increase, while S5 showed a sharp decrease in water content by d 228. Our interpretation of these data is that because we were unable to collect data between d 179 and 228, we likely missed measuring the maximum water content in S5 and may have missed it in S1 as well. Because S5 contains soils with high silt content, many gravel and small cobble layers, and numerous small voids, water infiltrated and drained through the profile more rapidly than in S1 where soils have higher clay content (Schwartz et al. 2008b).

Between d 204 (June 23, 2006) and d 282 (October 9, 2006), the rate of cumulative precipitation decreased and was again less than the cumulative PET slope (Figure 4, section III), indicating that any new precipitation

could be entirely removed from the system by ET. During this time period, water content decreased in all three profiles. Much of this decrease was the result of water removal by ET from the uppermost 1.5 m, though in S5 some of this decrease was also likely due to drainage through the bottom of the profile.

Depth-Dependent Changes in Soil Water

The timing of cumulative changes in volumetric soil water content (Figure 4) for each depth interval (Figure 2) helps to clarify the fate of soil water during the aforementioned wetting and drying periods. During the initial period of drying (d 137 to 173), there is evidence that some of the decrease in water content is due to loss through the bottom of the soil profile. For example, in profiles for S1 (Figures 4a and 4b), the interval between -1.5 and -3.0 m showed a slight decrease in water content between d 137 and 173. There is a corresponding increase in water below -4.5 m. Because both these intervals are

significantly below the depth at which ET would be a factor, we concluded that slow drainage was occurring in the soil profile. As some water drained from the -1.5 to -3.0 m interval, water content remained essentially constant between -3.0 and -4.5 m, and additional water was added below 4.5 m. The water is likely the tail end of a wetting front caused by rainfall prior to the first set of ERT measurements.

During the wetting interval between d 173 and 204 (Figure 4, section II), water infiltrated and drained to depths well below the influence of ET and could be assumed to contribute to potential recharge (Figures 4a through 4c). By d 179, a sharp increase in water content was measured in all three profiles in the depth interval -1.5 to -3.0 m: 6 d after the onset of heavy rain. By d 228, additional soil water was measured in S1 in the same depth interval, and water content also increased in the interval between -3.0 and -4.5 m. However, in S5 during the same time interval, water content significantly decreased between -1.5 and -3.0 m and sharply increased between -3.0 and -4.5 m in depth. This is seen as further evidence that soils in S5 allowed more rapid drainage than those in S1 and that the wetting front had moved to this depth in less than 55 d.

During the second drying interval (beginning at d 204), changes in ERT-derived water content for each 9-m-thick soil profile (Figure 4, section III) and analysis of changes occurring in individual depth intervals (Figure 4) showed that water losses primarily occurred in the upper 1.5 m of each profile, while deeper intervals showed the effects of a wetting front moving downward over time (Figure 4). This supported our hypothesis that water above -1.5 m would primarily be removed by ET during periods when PET was greater than precipitation, but water below -1.5 m would continue to move downward. Profiles in S1 display similar spatial and temporal patterns of soil water distribution. However, the profile in S5 shows different patterns that indicate that more rapid drainage was occurring.

Spatial and Temporal Variations in Soil Water

Figures 5 through 7 show the spatial distribution of increases or decreases in volumetric water content over time in each 2D profile. Temporally, there are two patterns of change that are evident. The first is a relatively uniform decrease in water content within the uppermost 1 to 3 m of the profiles between d 137 and 173. This was the result of removal by ET (while precipitation rates

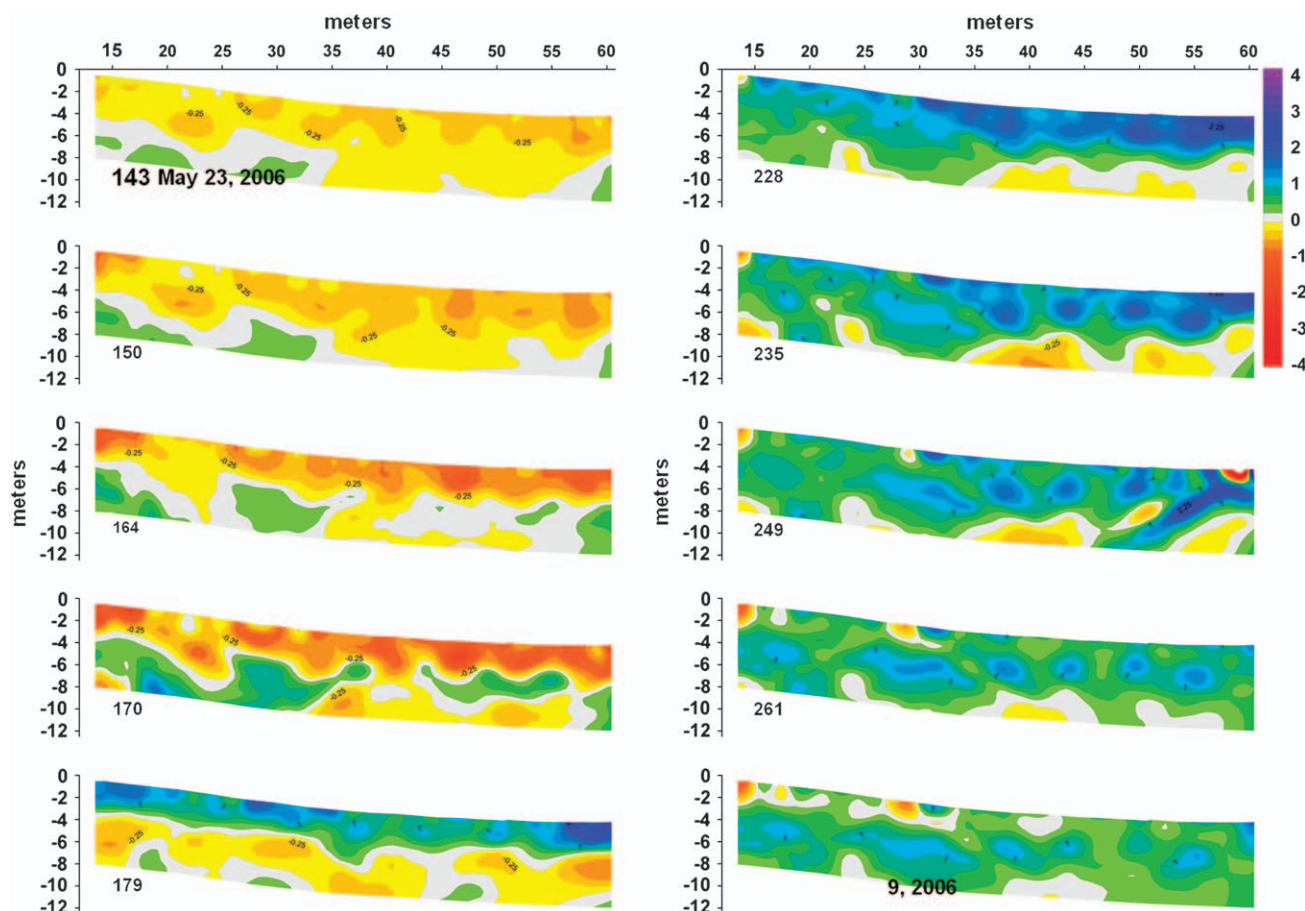


Figure 5. Temporal and spatial changes in volumetric soil water (% vol) relative to a baseline model on d 137 (May 17, 2006) in S1, transect #1 as modeled using time series ERT data. Warm colors represent decreases in volumetric water content and cooler colors represent increases in volumetric soil water content. Numbers in the lower left of each profile are Julian days. Large rain events occurred between d 173 and 187, with $PET < precipitation$ between d 173 and d 204.

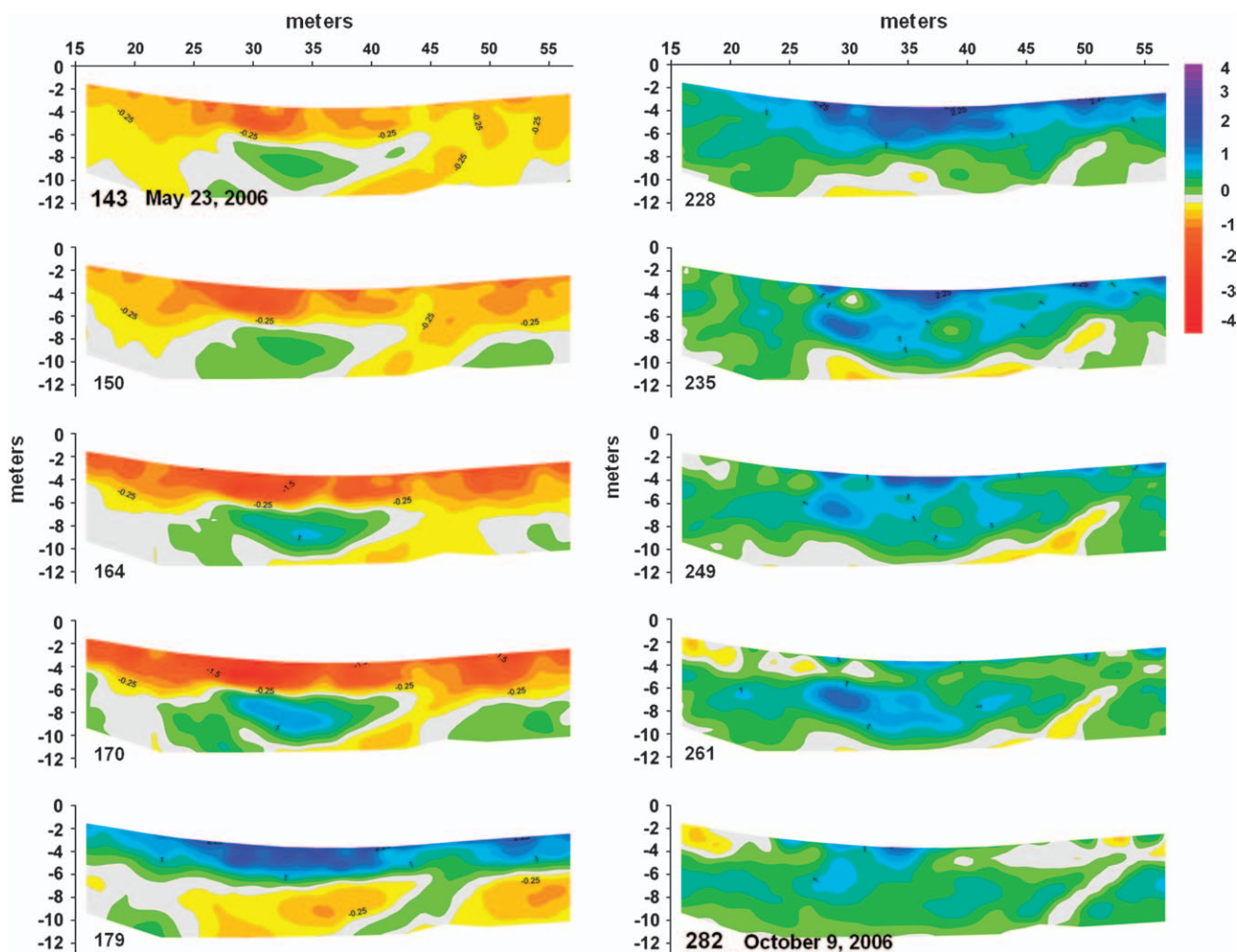


Figure 6. Temporal and spatial changes in volumetric soil water (% vol) relative to a baseline model on d 137 (May 17, 2006) in S1, transect #2 as modeled using time series ERT data. Warm colors represent decreases in volumetric water content and cooler colors represent increases in volumetric soil water content. Numbers in the lower left of each profile are Julian days. Large rain events occurred between d 173 and 187, with PET < precipitation between d 173 and 204.

were low) and slow drainage to deeper portions of the profile. In some cases, the latter resulted in slight increases in water content in deeper portions of the profile. The second pattern visible in Figures 5 through 7 is that large volumes of water infiltrated both on the flanks and in the bottom of the sinkholes after heavy rainfall between d 173 and 204. This created a large wetting front that moved downward and dispersed with time. Patterns of moisture change in Figures 5 through 7 also show large volumes of water infiltrated in the topographically lowest region of the sinkholes. Increased infiltration in the bottom of sinkholes was caused by runoff, which temporarily ponded after intense rainfall.

All three profiles show both an overall increase in water content along the entire profile length and large increases in the sinkhole bottoms. However, S5 does not show as much infiltration on the sinkhole flanks. Instead, it appears that more water infiltrated in the lowest portion of this sinkhole. Based on field measurements of 20 to 30 cm of ponded water in the lowest point of S5 immediately after a heavy rain, we believe ponding in this

sinkhole was greater than in S1. The amount of infiltration that occurs in the bottom of a sinkhole vs. the flanks is controlled by rates of precipitation (which would influence runoff), physical properties of the soils (which control rates of infiltration), type and amount of vegetation, and antecedent soil water conditions (which also influence how much and how fast infiltration occurs).

Potential Recharge

Using the aforementioned methods, we quantified potential recharge in each of the three 2D profiles (Figures 2 and 4) for the period between d 137 and 282 (May 17 to October 9, 2006). Figure 4 shows cumulative PET, precipitation, soil water content for the entire profile thickness, and soil water content for the portion of each profile below -1.5 m. By assuming that the maximum amount of soil water added to the interval below -1.5 m represents potential recharge, we estimated potential recharge resulting from infiltration and deep drainage during the rainy period between d 173 and 204 (June 22 to July 23, 2006) (Table 1). Our calculations reveal that

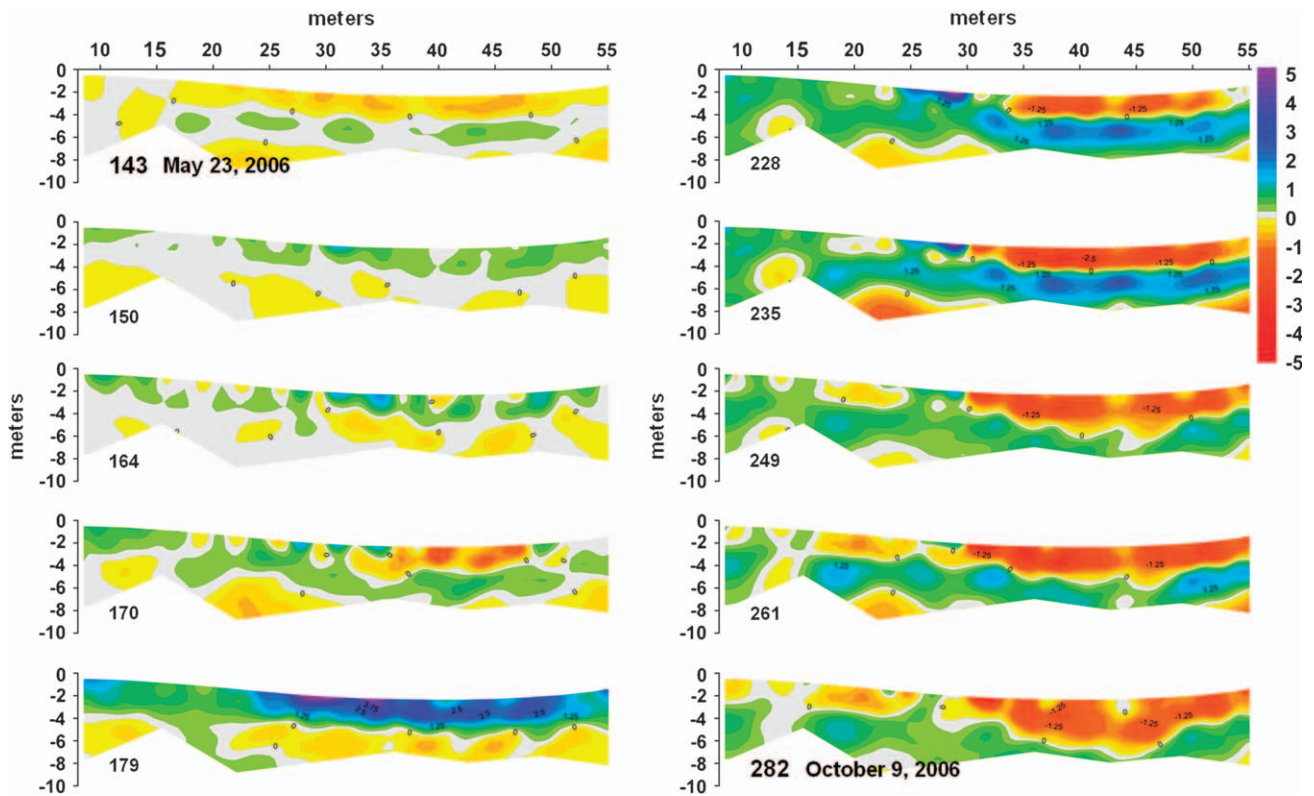


Figure 7. Temporal and spatial changes in volumetric soil water (% vol) relative to a baseline model on d 137 (May 17, 2006) in S5, transect #1 as modeled using time series ERT data. Warm colors represent decreases in volumetric water content and cooler colors represent increases in volumetric soil water content. Numbers in the lower left of each profile are Julian days. Large rain events occurred between d 173 and 187, with PET < precipitation between d 173 and 204.

between 19% and 31% of the precipitation, which fell between d 137 and 282, can be considered potential recharge. Differences in soil water content between the entire profile thickness vs. the portion below -1.5 m are assumed to represent soil water removed by ET (Figure 4). It is also possible that some of this water drained very rapidly through macropores in the soil and contributed to recharge without being detected by our ERT measurements. While there was some physical evidence that this was occurring in a few small root casts (we observed a rapid increase and decrease in access-tube TDR-measured soil moisture at two locations in S1 immediately after an intense storm), we believe this comprised a small percentage of the total recharge.

In Figures 4a and 4b (S1), soil water content below -1.5 m remained essentially constant after d 228. This was not the case in S5 where soil water content decreased rapidly after d 228 and even decreased to below what was measured at the beginning of the experiment (Figure 4c). These results are also visible in Figures 5 through 7.

Discussion

Our ERT-derived results compared well with the expected timing and amounts of infiltration and potential recharge using cumulative Penman-Monteith PET and precipitation data (Figures 4a through 4c). In all three profiles, the ERT-modeled increases in water content over

Table 1
Results of Recharge Calculations Showing Amount of Water Added to Each Profile below -1.5 m (derived from ERT data), Amount of Precipitation That Occurred over Each Profile, and Percent of Precipitation That Infiltrated below -1.5 m and Can Be Assumed to Represent Recharge

Sinkhole No. and Profile	Recharge (m^3)	Precipitation (m^3)	Recharge as % of Precipitation
#1, profile 1	5.1	17.4	29
#1, profile 2	4.7	15.1	31
#5, profile 1	3.2	17.1	19

Note: Time interval represented is between May 17 and October 9, 2006, but note that potential recharge occurred only over the time interval from d 173 to 204.

the entire profile thickness were consistent with nearly 100% infiltration during the intense rain events (section II). Total-profile water content decreased during the periods when rainfall rates were lower than modeled PET rates (sections I and III).

However, unusual patterns were observed (Figures 4b and 4c, section II) where the increase in total-profile ERT-derived water content was actually greater than the cumulative precipitation that fell over the profiles. We believe this is the result of runoff from the sinkhole flanks, which shed large volumes of water. Enhanced infiltration at the bottom of the sinkholes was measured in ERT profiles, which cross the bottom of the sinkhole. Evidence for the occurrence of runoff can be found in Figure 4a where the increase in ERT-derived water content is nearly identical to the precipitation that fell during the intense rain events beginning on d 173 but is significantly less than the infiltration measured in Figure 4b. The surface extent of the profile represented in Figure 4a ends at the bottom of the sinkhole. Thus, it does not show changes in water content across the entire width of the sinkhole bottom as is the case for the profile represented in Figure 4b.

Even though the two profiles in S1 suggest that different volumes of water infiltrated, both Figures 4a and 4b show that similar volumes of water were added to the depth interval below -1.5 m, with approximately 30% of cumulative precipitation contributing to potential recharge. When only compared to the interval where the rate of precipitation equaled or exceeded the rate of PET (d 173 to 204), approximately 50% of this precipitation contributed to potential recharge. Results from S5 were different from S1, though it appeared that a similar percentage of precipitation infiltrated and contributed to potential recharge below -1.5 m. The main difference between S5 and S1 was that the rate at which water passed through soils in S5 was much higher (Figure 7 vs. Figures 5 and 6). This is also evident from the temporal distribution of water content in the entire profile and at depths below -1.5 m (Figure 4c). After the rate of precipitation decreased to less than that of PET at d 204, near-surface soils lost water content rapidly (especially in silty soils in the lowest portion of the sinkhole). By d 282, there was almost no evidence of this water remaining.

Comparison with Other Recharge Measurements

Based on published estimates of annual recharge as a percent of precipitation in humid conditions similar to those found at our site (Delin and Risser 2007; Heppner et al. 2007; White and Burbey 2007; Nelms et al. 1997), the potential recharge values that we estimated using this ERT-based method are reasonable. In particular, Nelms et al. (1997) report that annual recharge ranges from 21.3 to 28 cm (8.4 to 11 inches) per year (median 24.6 cm, or 9.7 inches) in Virginia's Shenandoah Valley. This is approximately equivalent to a recharge rate of 25 percent of the mean annual precipitation in Blacksburg, Virginia, which is 104.1 cm (41 inches) (NOAA [National Oceanic and

Atmospheric Administration] 2008). It is worth pointing out that potential recharge was estimated during the summer growing season when PET was very high and recharge rates are normally very low and would thus not represent recharge rates during seasons with low PET. Recharge is extremely variable temporally and is dependent upon many different factors such as geology, climate, antecedent moisture conditions, physical and chemical soil parameters, and rates of precipitation and runoff (Delin et al. 2007; Delin and Risser 2007; Dripps 2003; Nolan et al. 2007).

Properties of Mantled Sinkholes

Our results show that mantled sinkholes contribute to recharge over long time scales (weeks to months) by temporarily storing and slowly releasing water that infiltrated and drained to depths below the influence of PET. The amount and rate of recharge depend on physical properties of the soils, as well as climatic conditions. Rather than characterizing all sinkholes as sources of rapid recharge, sinkholes should be considered within a hydrologic classification scheme in which one endmember is a sinkhole with no soil and an open conduit and the other is a sinkhole containing thick, clay-rich soils and no openings or macropores (a sinkhole pond). The sinkholes examined in this study lie closer to the latter in their characteristics (e.g., thick, clay-rich soils with a few observed root casts and burrows). It is, however, important to emphasize that all sinkholes, regardless of their overall hydrologic properties, should still be considered potential sources of both rapid and diffuse contamination. Even in a sinkhole dominated by slow drainage through thick soils, rapid flow and transport through macropores can still allow contaminants on the surface to rapidly reach the ground water.

Conclusions

Our results demonstrate that field-scale spatial and temporal distribution of soil water content can be estimated using ERT data. To do this, we used a modified form of Archie's law to convert 2D ERT data into 2D water content and then measured differences in modeled water content over time. From these data, we derived potential recharge amounts for each profile by assuming that water that drained below -1.5 m was potential recharge. After a short period of heavy rains, the maximum amount of soil water added below -1.5 m in the profiles was assumed to be the resulting potential recharge. These results compare well with patterns of cumulative PET and precipitation over the same time period. Our results also indicate that recharge rates in thickly mantled sinkholes are comparable to regional recharge rates, though the distribution of recharge through soil-mantled karst is temporally more diffuse than in most karst settings.

This work has demonstrated that mantled sinkholes are highly heterogeneous and can act as both rapid and slow sources of recharge to the underlying karst aquifers, refuting the assumptions that all sinkholes should be

treated simply as a source of rapid infiltration and recharge. These results underscore the need for better physical and hydrologic characterization of sinkholes.

One advantage of using this hydrogeophysical method to quantify potential recharge is that it is relatively easy to apply in long-term field-scale studies. After basic soil properties have been measured, the method is noninvasive and can be used over any time interval desired. There are often large discrepancies between results of small-scale recharge estimates or measurements and regional-scale results. For field-scale studies, the methods we present are an alternative to scaling large- or small-scale results down or up to obtain intermediate-scale estimates.

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